

## PLAYWRIGHT NOTES

### 1. What is Playwright?

- Playwright is an **open-source automation testing framework**.
- It is developed by **Microsoft**.
- It is mainly used for **automating and testing web applications**.
- It supports major browser engines:
  - **Chromium - chrome , edge**
  - **Firefox**
  - **WebKit - safari**
- It supports multiple programming languages such as:
  - JavaScript / TypeScript
  - Java
  - Python
  - .NET
- Playwright can be used for **end-to-end testing** of web applications.

### Interview Definition

Playwright is an open-source end-to-end test automation framework developed by Microsoft, used to automate and test web applications across Chromium, Firefox, and WebKit browsers.

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### 2. Why Do We Use Playwright?

We use Playwright to:

- Automate web application testing.
- Reduce **manual testing effort**.
- Test applications across different browsers.
- Execute tests faster.
- Test complete **end-to-end user workflows**.
- Handle modern and dynamic web applications.
- Run multiple tests in parallel.
- Identify defects early.
- Improve test coverage.
- Reduce repetitive testing work.

**Note : Flaky scripts : scripts might pass sometime and it might fail sometime(selenium cannot deal with dynamic elements(DOM Structure changes everytime) )**

## **Interview Answer**

We use Playwright to automate web application testing, reduce manual effort, perform cross-browser testing, improve test coverage, and execute repetitive and end-to-end test scenarios efficiently.

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### **3. Advantages of Playwright**

#### **1. Cross-Browser Support**

- Supports **Chromium, Firefox, and WebKit**.
- The same test can be executed across different browsers.

#### **2. Auto-Waiting → most import (30 sec)**

- Playwright automatically waits for elements to become ready before performing actions.
- Reduces synchronization issues(speed of execution and speed of the web app) and flaky tests.

#### **3. Fast Execution**

- Playwright is designed for fast and efficient test execution.

#### **4. Parallel Execution**

- Multiple tests can run simultaneously.
- Reduces overall execution time.

#### **5. Easy Browser Management**

- Supports browser, browser context, and page management.
- Allows us to create isolated browser sessions.

#### **6. Multiple Tab/Window Handling**

- Can easily handle multiple tabs, windows, and popups.

## **7. Powerful Debugging**

- Provides tools such as:
  - Playwright Inspector
  - Trace Viewer
  - Screenshots
  - Videos
  - HTML reports

## **8. Supports Modern Web Applications**

- Works well with dynamic and modern web applications.

## **9. Open Source**

- Playwright is freely available and has an active developer community.
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## **4. Features of Playwright**

Important features students should know:

- Cross-browser testing
  - Auto-waiting
  - Parallel test execution
  - Multiple tab and window handling
  - Iframe/Frame handling
  - Popup handling
  - Dialog handling
  - Mouse and keyboard actions
  - File upload and download
  - Screenshots and video recordings
  - Assertions
  - Hooks
  - Annotations
  - Test retries
  - Test timeout management - 30 sec
  - Trace Viewer
  - HTML reporting
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## 5. Playwright Test Runner → execution takes place

Playwright provides its own **built-in test runner**, called **Playwright Test**.

It provides features such as:

- Test execution
- Assertions
- Fixtures
- Hooks
- Parallel execution
- Annotations
- Reporting

### Interview Point

Playwright Test is the built-in test runner provided by Playwright for creating, executing, managing, and reporting automated tests.

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## 6. Playwright – Quick Interview Revision

### What is Playwright?

An open-source end-to-end test automation framework developed by Microsoft for testing web applications, api testing, mobile automation.

### Why do we use Playwright?

To automate web application testing, reduce manual effort, perform cross-browser testing, and execute tests efficiently.

### Which browsers does Playwright support?

Chromium, Firefox, and WebKit.

### Which languages does Playwright support?

JavaScript, TypeScript, Python, Java, and .NET.

### Is Playwright open source?

Yes, Playwright is an open-source framework developed by Microsoft.

## What is Playwright's biggest advantage?

It provides reliable cross-browser automation with features such as auto-waiting, parallel execution, browser contexts, powerful debugging.

## Installation:

1. **Install Node.js** – Playwright requires Node.js and npm.

**Why required?** Playwright's JavaScript/TypeScript tooling runs on **Node.js**.

**Use:** Provides the **runtime environment** to execute Playwright tests.

**What is it?** npm = **Node Package Manager**.

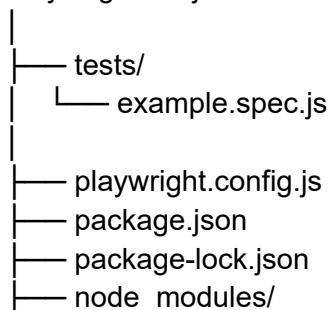
**Why required?** Used to **install and manage Playwright and other project dependencies**.

**Example:** `npm init playwright@latest`

2. **Install Visual Studio** – For writing test scripts
3. **Check installation** – Verify using `node -v` and `npm -v`.
4. **Create project folder** – Create and open the project in VS Code.
5. **Open Terminal** – Open the terminal inside the project folder.
6. **Initialize Playwright** – Run `npm init playwright@latest`.
7. **Select language** – Choose **JavaScript** or TypeScript.

## Playwright folder structure:

Playwright Project



### 1. `tests/`

- Main folder where we **store our test cases**.
- Usually contains files ending with `.spec.js` or `.test.js`.

**Example:** `login.spec.js`

### 2. `playwright.config.js`

- The **main configuration file** of Playwright.
  - Used to configure things like:
    - Browsers
    - Base URL
    - Timeout
    - Retries
    - Workers
    - Reports
    - Screenshots/videos
    - Projects
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### 3. **package.json**

- Contains the **project information and dependencies**.
  - Stores Playwright and other npm package details.
  - Also contains scripts used to run the project.
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### 4. **package-lock.json**

- Keeps the **exact versions of installed npm packages and their dependencies**.
  - Helps maintain consistent installations across environments.
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### 5. **node\_modules/**

- Contains the **installed packages/dependencies** required by the project.
- Created automatically when npm installs packages.

**Important:** We generally don't manually modify this folder.

### **Interview Point:**

The Playwright project structure mainly consists of the tests folder for test cases, playwright.config.js for configuration, package.json for dependencies and project scripts, node\_modules for installed packages, and report/result folders for test execution artifacts.

